

Test Plan — KAN-2: Automate Login Functionality

Status: DRAFT — pending human review **Source ticket:** [KAN-2](#) (AI_QA project)
Generated from: the actual KAN-2 ticket fetched from JIRA (2026-08-14)

1. Document Information

Field	Value
Project Name	AI_QA
Product/Application	VWO (Wingify) — Login page (subject to confirmation)
Version	TBD
Test Plan Version	0.1 (Draft)
Author	Test Plan Generator (skill)
Reviewer	TBD
Approval Date	TBD

2. Objective

Automate the login functionality using **Selenium with Python and Pytest**, per KAN-2. The automation must:

- Verify successful login with **valid credentials**.
- Verify **appropriate error messages** for **invalid credentials**.
- Add **assertions** to validate expected results.

The goal is a repeatable, CI-ready automated login test suite that catches regressions in the login flow.

3. Scope

In Scope

- Functional Testing (login happy path & error paths)
- UI automation (Selenium)
- Data-driven / parametrized credential tests
- Assertions on success redirect + error message content
- CI/CD execution readiness (headless browser)

Out of Scope

- Mobile / responsive testing
 - Performance / load testing
 - Security testing (penetration)
 - Non-login features
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4. Features to be Tested

- Login page loading
 - Valid credential login → successful navigation to dashboard/home
 - Invalid username → appropriate error message
 - Invalid password → appropriate error message
 - Empty fields → validation message (boundary)
 - Assertions validating expected vs. actual results
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5. Test Environment

Environment	URL	Purpose
QA	TBD	Primary automation target
UAT	TBD	Optional
Pre-Prod	TBD	Optional
Production	TBD	Not recommended for automation

Supported Platforms

- Windows (primary)
- macOS (CI parity)

Supported Browsers

- Chrome (primary)
 - Edge (optional)
 - Firefox (optional)
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6. Defect Management

Defect Lifecycle

New → Assigned → In Progress → Fixed → Retest → Closed (Reopened as needed)

Defect Tool

Jira (AI_QA project)

Severity

Critical / High / Medium / Low

Priority

P1 / P2 / P3 / P4

7. Test Strategy

Test Design Techniques

- Equivalence Partitioning (valid / invalid credential classes)
- Boundary Value Analysis (empty, single-char, max-length inputs)
- Error Guessing (wrong username, wrong password, both wrong)
- Use Case Testing (login as primary user journey)

Execution Strategy

1. Smoke test on a stable login page
2. Functional validation of each login scenario
3. Regression on login flow
4. CI run (headless)

Best Practices

- Shift-Left Testing
 - Risk-Based Testing
 - Continuous Testing (CI pipeline)
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8. Test Schedule

Activity	Planned Date	Owner	Status
Test Planning	TBD	QA Lead	Draft
Test Case Design	TBD	Test Engineer	Pending
Test Execution	TBD	Test Engineer	Pending
Regression	TBD	Test Engineer	Pending
Closure	TBD	QA Lead	Pending

9. Test Deliverables

- Test Plan (this document)
- Automated test suite (Python + Pytest + Selenium)
- Test Execution Report
- Defect Report
- Test Summary Report
- Sign-off Document

10. Test Scenarios (P0/P1/P2)

Each scenario maps to an explicit requirement in the ticket or a gap identified below.

ID	Scenario	Expected Result	Priority	Maps To
TC-01	Login page loads successfully	Page title/URL correct, login form visible	P0	AC: verify login
TC-02	Valid credentials → login	Successful navigation to dashboard/home	P0	AC: valid login
TC-03	Invalid username	Appropriate error message displayed	P0	AC: invalid creds
TC-04	Invalid password	Appropriate error message displayed	P0	AC: invalid creds
TC-05	Both username & password invalid	Appropriate error message displayed	P1	Gap: negative path
TC-06	Empty username field	Validation/error message displayed	P1	Gap: boundary
TC-07	Empty password field	Validation/error message displayed	P1	Gap: boundary

ID	Scenario	Expected Result	Priority	Maps To
TC-08	Whitespace-only input	Treated as empty or error message	P1	Gap: boundary
TC-09	Case-sensitive username/password	Mismatched case → error message	P1	Gap: ambiguity
TC-10	Assertions validate all expected results	No false positives; each assertion maps to expected state	P0	AC: assertions
TC-11	Locked/disabled account (if applicable)	Clear lockout message	P2	Gap: security
TC-12	Session/cookie set after login	Session persists; logout works	P2	Gap: state

11. Test Data & Environment

Test Data

Data	Value	Notes
Valid username	TBD (from test env)	Must be a real active account
Valid password	TBD (from test env / secrets)	Store securely, never in repo
Invalid username	TBD	e.g. <code>invalid_user@test.com</code>
Invalid password	TBD	e.g. <code>WrongPass!1</code>

Environment

- Base URL of the login page (TBD)
- Browser: Chrome (headless for CI)
- WebDriver: Selenium WebDriver matching browser version

12. Entry & Exit Criteria

Entry

- Stable login page available at known URL
- Valid test credentials provisioned
- Selenium/Pytest environment ready

Exit

- All P0 scenarios pass

- All assertions validated
- Defects (if any) logged in Jira
- Test execution report produced

13. Risks & Assumptions

Assumptions

- The login page under test is the VWO (Wingify) login at the app URL (to be confirmed by author).
- Valid credentials will be provided by the ticket owner (not included in the ticket).
- Browser automation (Selenium) is permitted on the target environment.
- The error message text for invalid credentials is deterministic and assertable.

Risks

Risk	Mitigation
No valid credentials provided	Blocking — request from ticket author
Target URL not confirmed	Blocking — confirm which login page
Error message wording varies by locale	Pin exact expected strings after first run
CAPTCHA / 2FA on login	Use test environment with CAPTCHA/2FA disabled or handled
Headless browser differences	Run smoke locally + headless in CI
Credentials stored insecurely	Use env vars / secrets manager, never commit

14. Human Review Gate

Do not proceed to writing test cases or automation until a human approves.

Assumptions made

- Target app is the VWO (Wingify) login page — **not stated in the ticket.**
- Valid test credentials will be supplied externally — **not in the ticket.**
- "Appropriate error messages" means deterministic, assertable strings.

Open questions for the ticket author (blocking)

1. **Which application/login URL** should the automation target? (VWO/Wingify app URL?)
2. **What are the valid test credentials** (and where should they be stored)?
3. What **exact error message text** is expected for invalid credentials (username vs password)?
4. Is there **CAPTCHA / 2FA / SSO** on the login that the automation must handle or bypass?
5. Which **browsers/platforms** must the suite support (Chrome only, or Edge/Firefox too)?
6. Should the suite run in **CI (headless)** — and if so, which pipeline?

Required before sign-off

- ☐ Confirm target login URL
- ☐ Provide valid test credentials
- ☐ Confirm expected error-message strings
- ☐ Confirm browser/platform matrix
- ☐ Approve test scenarios (P0/P1/P2)